FUTURE TRENDS AI

AUTOMOTIVE INDUSTRY



By **Gopalakrishnan C.S.**

THE THINKING FACTORY OF TOMORROW

WITH AI EMBEDDED ACROSS PLANNING, PRODUCTION AND QUALITY, AUTO COMPANIES ARE MAKING INTELLIGENT MANUFACTURING THE FOUNDATION OF INDUSTRIAL RESILIENCE

Artificial Intelligence is no longer a bolt-on upgrade to manufacturing—it is the operating system of modern industry. Global thought leaders converge on this point: PwC frames the shift as moving from "doing digital" to "being digital", embedding intelligence into the very fabric of workflows; McKinsey underscores that productivity gains now hinge on human-machine partnerships; Deloitte positions Industry 4.0 as a resilience strategy, wherein smart factories and intelligent applications are foundational to competitiveness, not optional extras. India is well placed to lead this transformation.

AI & Smart Manufacturing

At Hyundai Motor India Limited (HMIL), technology drives every aspect of operations through a data-centric, intelligent ecosystem. Our IT landscape integrates advanced digital infrastructure with streamlined processes, spanning the entire value chain—from production planning and state-of-the-art manufacturing to digital-first customer engagement.

Predictive analytics and real-time monitoring empower engineering teams to optimise equipment maintenance and process control. By continuously tracking critical parameters and applying AI models, we resolve bottlenecks before they become downtime. Quality assurance is equally transformed: digital pre-assembly, virtual simulations and AI vision systems ensure dimensional precision, validate torque specifications and rigorously test electronics—all culminating in structured sign-offs that protect our promise of uncompromising build quality.

Our Software Defined Factory (SDF) vision orchestrates production end-to-end by converging data, connectivity and intelligence. Over 2,000 critical machines generate